4 **(a)** State what is meant by an electric field.

……...

……... [1]

(b) Fig. 4.1 below shows a set of equipotential lines of a region of an electric field.

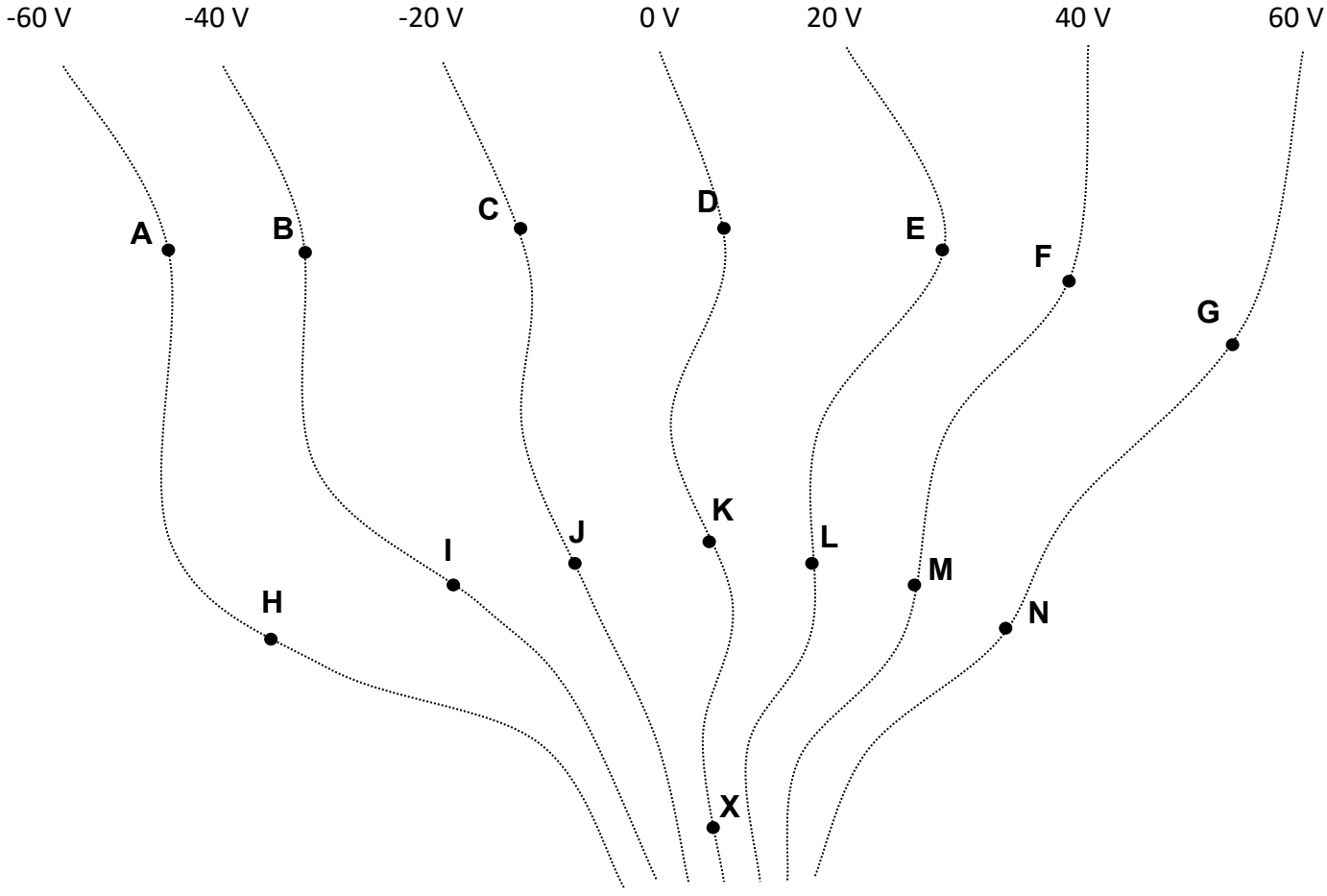

Fig. 4.1 (drawn to scale)

(i) Given that the same field line passes through the points **A,B,C,D,E,F** and **G,** draw in Fig. 4.1, this field line, clearly indicating the direction of the field. [2]

(ii) Explain whether the field is stronger at **D** or at **X**.

……

…… [1]

(iii) A charge Q_1 of - 5.0 μC is placed at the point **L**. Calculate the electric potential energy of the charge Q_1 at point **L**.

electric potential energy = …….. J [2]

(iv) Charge Q_1 is now released from rest. Given that work done on it *by the electric field* is 1.0×10^{-4} J, identify the point(s) that can represent the final location of Q_1.

Point(s): ……. [2]

(v) Charge Q_1 is now removed, and Q_2 of - 15.0 μC is now placed at the point **X**. By making *direct measurements* from the Fig 4.1, determine the electric force experienced by the charge.

Indicate on Fig. 4.1 the direction of the force experienced by Q_2 at **X**.

electric force = ……. N [3]